

BELIMO

Energy Valve Coil Estimator

(Belimo Energy Valve Coil Estimator.xlsm)

USER MANUAL

Release 4.3

Prepared for Belimo internal staff – not for distribution.

Belimo U.S., Inc. | 33 Turner Road, Danbury, CT 06810

Languages: English · Español · Português · Français | Units: US or Metric

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1. Introduction

The Belimo Energy Valve Coil Estimator (the “tool”, file name Belimo Energy Valve Coil Estimator.xlsm) is an Excel workbook that answers one question about one air handler: *“So... how much will this smart valve save me on this coil?”* You enter the coil’s design data, how long it runs, where the building is, and the Delta-T the coil delivers today and the Delta-T an Energy Valve will hold. The tool works out how much less water the pump has to move, what that saves in energy and money, and what the investment is worth over 5, 10, 15 and 20 years.

More flow costs money, and it costs it disproportionately: pump power varies with roughly the cube of flow, so cutting flow by a third cuts pump power by around 70%. An Energy Valve holds the coil at the Delta-T you ask for, so the flow – and the pumping bill – comes down.

Scope of the tool

This is a **coil-level** tool and it counts **pumping savings only**. Its companion, the Energy Valve Savings Estimator, models a whole chilled-water plant and adds the chiller-side savings. Use this tool when the conversation is about a specific coil or air handler; use the plant tool when it is about the central plant. It is an estimating tool for internal use. Results depend on the accuracy of the data entered and do not take external factors into account.

1.1 How the workbook is laid out

Tab	What it is for
Home	Everything you fill in, in five numbered steps, plus the headline financial results.
Pump Savings	The calculation engine: one block per Delta-T, plus Energy Valve model selection.
Flow Curves	Reference curves relating the control signal to flow through the valve.
Charts	The same relationships plotted.
Results	The customer-facing summary. This is the page you present.

1.2 What’s in this manual

- Section 2 – Getting started: opening the file, the About screen, updates, which cells you can type in, language and units.
- Section 3 – The Home tab, step by step.
- Section 4 – The Pump Savings tab.
- Section 5 – Flow Curves and Charts.
- Section 6 – The Results tab.

- Section 7 – The financial results explained.
- Section 8 – How the numbers are calculated (for anyone who wants to check the arithmetic).
- Section 9 – Practicalities and troubleshooting. Appendix – Release history.

2. Getting Started

2.1 Opening the file

Open the workbook in Excel for Windows. It contains macros, so if Excel shows a yellow **Protected View** bar, click **Enable Editing**, and if it then shows a **Security Warning** bar, click **Enable Content**. (For a file that came from e-mail or a download, right-click it in File Explorer, choose Properties, tick **Unblock** and click OK first.)

On opening, the tool checks Belimo's update source for a newer release (the first time a new one appears it offers to install it), takes you to the Home tab, and re-applies your saved language. The workbook opens with a worked example already filled in (a 310 GPM coil in the North East) so you can see every result before you enter your own data. Overwrite the example values in Steps 2 to 5 with your own.

2.2 The About screen

Click the small round **(i)** button at the top right of the Home tab (beside the Step 1 callout) to open the About screen. It shows the tool's name, version and release date, a short description, and who maintains the tool.

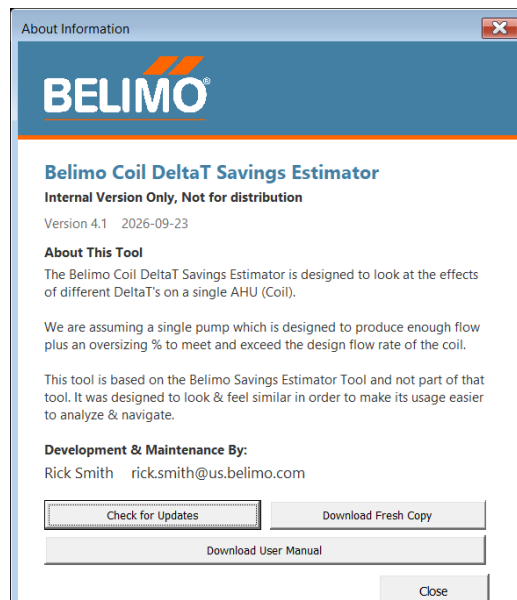


Figure 1 – The About screen.

Button	What it does
Check for Updates	Asks the update source whether a newer release exists. If one does, you are told what is new and may choose to download it.
Download Fresh Copy	Downloads the current published release straight away, even if you are already up to date – useful if you think your copy is damaged.

Button	What it does
Download User Manual	Saves the current published copy of this manual (a PDF) to a folder you choose. If none has been published yet, the tool tells you so.
Close	Closes the About screen.

When a newer release is published the tool tells you the next time you open it, once for each new version; Check for Updates on the About screen does the same at any time and offers a summary of what changed. If you choose to install it, you pick where to save the new copy through a normal Windows **Save As** dialog. Your current copy is never changed or deleted. If the update source cannot be reached, the check does nothing and the tool works as normal.

2.3 The step callouts

The Home tab is laid out as a wizard. Each orange section band has a numbered blue **Step** callout to its right telling you what to enter in that block. Work top to bottom and the tool takes care of the rest.

2.4 Which cells can I type in?

Shading	Meaning
Pale / near-white	Editable. Type here. (The key “Editable Field” / “Label” is shown on the sheets.)
Blue-grey	A label or a calculated result. Protected – the tool writes it.
Orange band	A section heading.

The sheets are protected so a stray keystroke cannot break a formula. If a cell will not accept typing, that is the tool telling you it is a result rather than an input.

2.5 Language and units

Control	Cell	Choices and effect
Language	F5	English, Español, Português or Français. Translates every label, heading, unit word and help message on every tab.
Units	F6	US or Metric. Switches every flow, head, temperature, power and volume figure and its unit.

The calculation always runs in US units internally and converts for display, so switching back and forth never drifts. Figure 2 shows the same Home tab in Spanish with Metric units. Note that the values you have already entered are converted, not re-typed: 310 GPM becomes 1173.478 L/min and a 12 °F Delta-T becomes 6.667 K (shown as °C).

Belimo Energy Valve
Estimador de Ahorros ΔT de Serpentina

Selección de Idioma y Unidades

Idioma

Unidades

Español
Metric

▼

Datos de Entrada de Diseño de la Serpentina

	Valor	Unidad
Costo de Electricidad (Promedio)	0.10	\$/kWh
Carga de Diseño de la Serpentina	1173.478	LPM
Número de Serpentin	1	uds.
Horas de Operación / Día	20	horas
Días de Operación / Semana	6	días
Semanas de Operación / Año	52	semanas
Altura Real de la Bomba Primaria (Diseño)	89.672	kPa
Eficiencia de Bomba, VFD, Motor (Promedio)	70	%
Delta T de Diseño	6.667	°C
Delta T Real (Antes de la Instalación de Energy Valve)	3.333	°C
Delta T Futuro (Configuración del Gestor de Delta T de Energy Valve)	6.667	°C

Datos de Entrada de Tiempo de Operación

Horas de Operación / Día	20	horas
Días de Operación / Semar	6	días
Semanas de Operación / A	52	semanas
Horas de Operación Anua	5497	horas

Campo Editable

Etiqueta

Paso 1

Selecione su Idioma y sistema de Unidades preferido

Paso 2

Ingrese en los recuadros de la izquierda, lo siguiente:
Costos Eléctricos
GPM de Diseño de la Serpentina
Horas de Operación
Altura de la Bomba

Paso 3

Ingrese en los recuadros de la izquierda, lo siguiente:

Figure 2 – The top of the Home tab in Spanish with Metric units.

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3. The Home Tab – the Five Steps

Everything you type lives on the Home tab. Figure 3 shows Steps 1 and 2.

3.1 Steps 1–2: Language and coil design data

The screenshot shows the 'Belimo Energy Valve Coil ΔT Savings Estimator' interface. It is divided into three main sections: 'Language & Unit Selection', 'Coil Design Input Data', and 'Operating Time Input Data'. On the right, there are three steps: Step 1 (Language & Unit Selection), Step 2 (Coil Design Input Data), and Step 3 (Operating Time Input Data). Step 1 is currently active, showing a dropdown for 'Language' set to 'English' and 'Units' set to 'US'. Step 2 is also active, showing a table of input data with example values. Step 3 is inactive, showing a dropdown for 'Climate Region'.

Language & Unit Selection	
Language	English
Units	US

Coil Design Input Data		
Electricity Cost (Average)	0.10	\$/kWh
Design Coil Load	310	GPM
Number Of Coils	1	pcs.
Operating Hours / Day	20	hours
Operating Days / Week	6	days
Operating Weeks / Year	52	weeks
Actual Primary Pump Head (Design)	30	ft.
Pump, VFD, Motor Efficiency (Average)	70	%
Design Delta T	12	°F
Actual Delta T (Prior To Energy Valve Installation)	6	°F
Future Delta T (Energy Valve Delta T Manager Setting)	12	°F

Operating Time Input Data		
Operating Hours / Day	20	hours
Operating Days / Week	6	days
Operating Weeks / Year	52	weeks
Annual Operating Hours	5497	hours

Figure 3 – Steps 1 and 2 – language and units, and the coil design data (with the example values).

Step 1 sets language and units (Section 2.5). Step 2 is the coil itself – the values every later number is built from.

Field	Cell	Unit	What it is
Electricity Cost (Average)	F10	\$/kWh	Blended all-in rate. Every dollar figure in the workbook runs through this one number.
Design Coil Load	F11	GPM	The coil's design flow . Despite the word "Load", this is a flow rate in gallons per minute (or L/min in Metric), not a tonnage – note the unit beside it.
Number Of Coils	F12	pcs.	Fixed at 1 – this tool estimates one coil at a time. (Its help icon notes multiple coils as a future release.)
Operating Hours / Day	F13	hours	With the two below, sets annual run hours.
Operating Days / Week	F14	days	
Operating Weeks / Year	F15	weeks	
Actual Primary Pump Head (Design)	F16	ft.	Head the pump works against at design flow (kPa in Metric).

Field	Cell	Unit	What it is
Pump, VFD, Motor Efficiency (Average)	F17	%	Combined wire-to-water efficiency. If you do not know it, the help icon suggests 70–80%.
Design Delta T	F18	°F	What the coil was selected for.
Actual Delta T (Prior To Energy Valve Installation)	F19	°F	What it achieves today, before the Energy Valve – the “before” case.
Future Delta T (Energy Valve Delta T Manager Setting)	F20	°F	What the Energy Valve’s Delta-T Manager will hold – the “after” case.

The two Delta-T fields at the bottom are the whole estimate. Actual and Future are the before-and-after the entire comparison is built from, and the tool enforces a minimum gap of 2 °F between them (Future must be at least 2 degrees above Actual) so the pair always makes physical sense; a value outside the allowed range is refused when you type it.

Delta-Ts snap to six fixed scenarios: 4, 6, 8, 10, 12 and 14 °F

The Results tab compares six fixed Delta-T scenarios: **4, 6, 8, 10, 12 and 14 °F** (in Metric the same six steps, for example 3.33 for 6 °F). Actual and Future must be one of them. If you type another value, the tool adjusts it to the nearest scenario and tells you: 7 becomes 8, 13 becomes 14, and an Actual below 4 becomes 4. Actual can be at most 12 and Future at least 6, and Future is always at least 2 degrees above Actual. The adjusted values are what appear in the box and on the Results tab.

3.2 Steps 3–4: Operating time and climate region

Annual Operating Hours 5497 hours
Climate Region North East

Point	Bin	Load	% of Operating Time	Operating Hours
A	Design	100%	25.0%	1374
B	Peak	75%	50.0%	2748
C	Low	50%	25.0%	1374
D	Min	25%	0.0%	0
			100%	5497

Climate Region Degree Day Input Data

Climate Region	Cooling Degree Days	Humidity Adjustment	Multiplier
North East	1034	30%	0.88
Central	1090	30%	0.93
East North Central	975	30%	0.83
West North Central	1123	15%	0.85
South East	2264	30%	1.93
South	2004	30%	1.71
South West	2750	5%	1.89
Mountain	1450	5%	1.00
North West	720	5%	0.50
West	1850	5%	1.27
Custom Region	800	0%	0.52
Average US	1526	19%	1.19

Selected Value 0.88

Climate region from the drop down box

% of Operating Time
Enter Percentages of operating time.

Step 4

Verify your Climate Region or select custom and insert your own value.

Figure 4 – Steps 3 and 4 – operating time, load points and climate region.

Annual operating hours are computed from your three entries and then adjusted for climate: hours per day × days per week × weeks per year × the **climate multiplier** for your region (the Selected Value

beneath the region table). With the example values, $20 \times 6 \times 52 = 6,240$ hours $\times 0.88$ (North East) = 5,497 hours.

Rather than simulate every hour of the year, the tool uses four representative operating points and gives each a share of the running time. Only the shares for points A, B and C are editable; point D is calculated as whatever is left. Keep A + B + C at 100% or less: the total cell shows “Error, must = 100%” if they add up to more than 100%.

Point	Cell	Load	Default share	Meaning
A – Design	E33	100%	25%	Full-load hours.
B – Peak	E34	75%	50%	The bulk of a typical year.
C – Low	E35	50%	25%	Light load.
D – Min	E36	25%	0%	Not editable – calculated as whatever is left after A, B and C (0% when they total 100%). Its hours are included in the annual total.

Climate Region (cell E30) is a drop-down. Each region supplies a cooling-degree-day figure and a humidity adjustment, from which the multiplier is calculated relative to the average US region:

Region	Cooling degree days	Humidity adjustment	Multiplier
North East	1034	30%	0.88
Central	1090	30%	0.93
East North Central	975	30%	0.83
West North Central	1123	15%	0.85
South East	2264	30%	1.93
South	2004	30%	1.71
South West	2750	5%	1.89
Mountain	1450	5%	1.00
North West	720	5%	0.50
West	1850	5%	1.27
Custom Region	800 (editable)	0% (editable)	0.52
Average US	1526	19%	1.19

For a site that fits none of the standard regions, type your own cooling degree days and humidity adjustment in the **Custom Region** row (cells E53 and F53), then choose “Custom Region” in the drop-down. The multiplier is (region degree days $\div$ average degree days) $\times$ (1 + humidity adjustment).

3.3 Step 5: Financial inputs and results

Financial Inputs

Step 5

Investment Cost (Energy Valve + Installation)	\$4,000.00	\$			
Discount Rate	4.0%	%			
Estimated Measure Life	15	years			
	5 years	10 years	15 years	20 years	
Net Present Value (NPV)	\$3,488	\$9,643	\$14,701	\$18,859	
Savings-Investment Ratio (SIR)	1.87	3.41	4.68	5.71	
Simple Payback Period	2.4 years				

Disclaimer

Enter the estimated cost to purchase Energy Valves and the cost of installation.

Figure 5 – Step 5 – financial inputs and the three financial results at four horizons.

Field	Cell	Unit	What it is
Investment Cost (Energy Valve + Installation)	F62	\$	Use the incremental cost over the valve that would otherwise have been fitted, not the whole valve package.
Discount Rate	F63	%	Converts future dollars into today's. Usually the organisation's cost of capital or required return. 3–6% is common for public and utility-funded work.
Estimated Measure Life	F64	years	How long the measure is expected to keep saving. Feeds the lifetime figures on the Results tab.

Below the inputs, three results appear at 5, 10, 15 and 20 years: the **Net Present Value (NPV)**, the **Savings-Investment Ratio (SIR)** and the **Simple Payback Period**. All three are explained in Section 7. The small (i) icons beside the Discount Rate, NPV and SIR open a plain-language explanation of each.

3.4 The help icons and the Disclaimer

Every round blue (i) icon opens a short help message in your chosen language. On the Home tab they explain Number of Coils, Pump Efficiency, Discount Rate, NPV and SIR; other tabs have their own (Sections 4 and 6). Click Close to dismiss.

A **Disclaimer** band at the foot of the Home tab states that the tool is not copyrighted by Belimo Aircontrols (USA), Inc., is strictly an internal estimating tool, that results depend on the accuracy of the data entered and do not consider external factors, and that it looks at the effects of pump costs over various Delta-Ts for a single coil using standard pump affinity rules.

4. The Pump Savings Tab

The engine. Nothing here needs filling in except one factor – but it is worth understanding what it is doing.

Belimo Energy Valve

Coil ΔT Savings Estimator

Coil Design Data

Pump Design Factor	125%	Max Flow Rate	387.5 GPM	Editable Field
Minimum Pump head	0			Label
Design Chilled Water Coil Flow [V'max]	310			
Design Chilled Water Coil ΔT	12			
Design Chilled Water Coil [tons]	155			
Design Chilled Water Coil [BTU/h]	1,860,000	Energy Valve Model	EV400SU-317	95.317 gpm 4" DN100 * based on V'max

Coil Design Performance Data

Load [%]	Time [%]	Time [h]	Flow [GPM]	Head [ft.]	Power [HP]	BTU/h	Energy [kWh]	Annual gallons	Pump Costs	Pump costs	
100	25.0	1374	310	30	3.4	1,860,000	3,439	25,559,179	\$344	Annual gallons Avg GPM Energy [kWh]	
75	50.0	2748	233	17	1.4	1,395,000	2,902	38,338,769	\$290		
50	25.0	1374	155	8	0.4	930,000	430	12,779,590	\$43		
25	0.0	0	78	2	0.1	465,000	0	0	\$0		
									6,771	76,677,537	677

Chilled Water Coil Flows at different Delta T [°F]

Chilled Water Coil Flow at Delta T [°F]: 4

Load [%]	Time [%]	Time [h]	Flow [GPM]	Head [ft.]	Power [HP]	BTU/h	Energy [kWh]	Annual gallons	BTU/h	Increase
100	25.0	1374	388	47	6.6	775,000	6,717	31,948,974	42%	78
75	50.0	2748	388	47	6.6	775,000	13,435	63,897,948	56%	155
50	25.0	1374	388	47	6.6	775,000	6,717	31,948,974	83%	233
25	0.0	0	233	17	1.4	465,000	0	0	100%	155

26,869 127,795,895

Chilled Water Coil Flow at Delta T [°F]: 6

Load [%]	Time [%]	Time [h]	Flow [GPM]	Head [ft.]	Power [HP]	BTU/h	Energy [kWh]	Annual gallons	BTU/h	Increase
100	25.0	1374	388	47	6.6	775,000	6,717	31,948,974	42%	78
75	50.0	2748	388	47	6.6	775,000	13,435	63,897,948	56%	155
50	25.0	1374	388	47	6.6	775,000	6,717	31,948,974	83%	233
25	0.0	0	233	17	1.4	465,000	0	0	100%	155

Figure 6 – The Pump Savings tab: design data, valve selection, the design-point performance table and the first Delta-T block.

4.1 Design data

Field	Cell	What it is
Pump Design Factor	F5	The only editable cell on this tab (default 125%). A safety margin on design flow – how much bigger the pump is than the coil strictly needs. It caps how much flow the pump can deliver (the Max Flow Rate beside it is design flow × this factor). Leave it alone unless you know the real oversizing.
Minimum Pump head	F6	Reserved for a future release; leave at 0.
Design Chilled Water Coil Flow [V'max]	F7	Your Design Coil Load from the Home tab.
Design Chilled Water Coil ΔT	F8	Your Design Delta T.
Design Chilled Water Coil [tons] / [BTU/h]	F9, F10	The design cooling capacity implied by the flow and Delta-T (tons = GPM × ΔT ÷ 24; BTU/h = GPM × ΔT × 500).

4.2 Energy Valve selection

Using the design flow, the tool picks the Energy Valve model that covers it and shows the model, its flow range and its pipe size beside the design data (for example EV400SU-317, 95–317 gpm, 4" DN100). The choice is based on V_{max}, the design flow. The range is shown in whichever units you selected.

Design flow (GPM) from	Model	Flow range	Pipe size
1.65	EV050	1.65–6.6 gpm	0.5" DN15
6.6	EV075	2.7–11 gpm	0.75" DN20
12	EV100	4.5–18.2 gpm	1" DN25
19	EV125	7.1–28.5 gpm	1.25" DN32
29	EV150	11–44 gpm	1.5" DN40
45	EV200	16.5–66 gpm	2" DN50
67	EV200H	25–100 gpm	2" DN50
101	EV250SU-127	38–127 gpm	2.5" DN65
128	EV300SU-180	54–180 gpm	3" DN80
181	EV400SU-317	95–317 gpm	4" DN100
318	EV500SU-495	149–495 gpm	5" DN125
496	EV600SU-713	214–713 gpm	6" DN150
714	Dual EV's	Up to 1400 gpm	–

The selection is a guide for the conversation, not a sizing calculation; confirm the valve with a full sizing before quoting.

4.3 The performance blocks

Below the design data, the tab has a block for the design case and then one block for each Delta-T scenario (4, 6, 8, 10, 12 and 14 °F). Each block carries the four load points across flow, head, power, energy and cost, and they all work the same way: for each load point the tool works out the flow needed at that Delta-T, the head the pump sees at that flow, the resulting pump power and, from the hours in that bin, the energy, water volume and cost. Add the four rows and you have that Delta-T's annual pumping bill. Section 8 gives the formulas.

4.4 Saving the tab as a PDF

The picture button at the top right of the tab saves the sheet you are on as a PDF. It offers a file name that includes the sheet name and the date and time, in the workbook's folder, through a Save As dialog, and confirms with the file location when done.

Help icons on this tab

The (i) beside Pump Design Factor explains it as the percentage of additional pumping available due to over-sizing or future design. The (i) beside Minimum Pump head is marked as a future feature.

5. Flow Curves and Charts

Reference material. Nothing to fill in on either tab. These tabs are there for the technical conversation – how a control signal maps to valve position and flow, and what that looks like plotted. They are useful when a customer or controls contractor asks how the valve behaves across its range; they are **not** part of the savings calculation.

5.1 Flow Curves

Flow Curves relates the DDC control signal to the actual flow through the valve, at the design flow you entered. Down the left, each row is one step of valve opening (0–100%) with the corresponding control voltage; the columns show the flow for a linear characteristic and for an equal-percentage characteristic, and the heat transfer (BTU/h) at the design Delta-T. The blue columns are the ones to use, and a worked example is printed at the right: an Energy Valve with a V'max of 310 GPM at 2.88 volts flows 17.67 GPM at 6% valve opening, producing 106,023 BTU/h at a 12 °F Delta-T (this example text is fixed and does not change with your inputs). The right-hand block feeds the charts.

BELIMO® Belimo Energy Valve									
Used to Calculate Projected Flow									
DDC Control signal		Valve Position & Flow based on DDC Voltage							
Voltage	Voltage	Linear Part	Equal Part	Vmax	Vmax	ΔT	btu/h		
0.50	2.00	Linear	EQ %	Linear	EQ %				
	10.00			310	310	12			
In %	In V	In V	In %	In %	In %	In %	btu/h		
0	0.50	2.00	0.0%	4.0%	0.00	0.00	0		
1	0.60	2.08	0.4%	4.1%	0.00	0.00	0		
2	0.69	2.16	0.7%	4.3%	0.00	0.00	0		
3	0.79	2.24	1.1%	4.4%	0.00	0.00	0		
4	0.88	2.32	1.4%	4.5%	0.00	0.00	0		
5	0.98	2.40	1.8%	4.7%	0.00	0.00	0		
6	1.07	2.48	2.1%	4.9%	0.00	0.00	0		
7	1.17	2.56	2.5%	5.0%	0.00	15.54	93,210		
8	1.26	2.64	2.8%	5.2%	8.68	16.04	96,260		
9	1.36	2.72	3.2%	5.3%	9.77	16.57	99,410		
10	1.45	2.80	3.5%	5.5%	10.85	17.11	102,663		
11	1.55	2.88	3.9%	5.7%	11.94	17.67	106,023		
12	1.64	2.96	4.2%	5.9%	13.02	18.25	109,492		
13	1.74	3.04	4.6%	6.1%	14.11	18.85	113,076		
14	1.83	3.12	4.9%	6.3%	15.19	19.46	116,776		
15	1.93	3.20	5.3%	6.5%	16.28	20.10	120,597		
16	2.02	3.28	5.6%	6.7%	17.36	20.76	124,544		
17	2.12	3.36	6.0%	6.9%	18.45	21.44	128,619		
18	2.21	3.44	6.3%	7.1%	19.53	22.14	132,828		
19	2.31	3.52	6.7%	7.4%	20.62	22.86	137,175		
20	2.40	3.60	7.0%	7.6%	21.70	23.61	141,664		
21	2.50	3.68	7.4%	7.9%	22.79	24.38	146,300		
22	2.59	3.76	7.7%	8.1%	23.87	25.18	151,087		
23	2.69	3.84	8.1%	8.4%	24.96	26.01	156,031		
24	2.78	3.92	8.4%	8.7%	26.04	26.86	161,137		
25	2.88	4.00	8.8%	8.9%	27.13	27.74	166,410		
26	2.97	4.08	9.1%	9.2%	28.21	28.64	171,856		
27	3.07	4.16	9.5%	9.5%	29.30	29.58	177,480		
28	3.16	4.24	9.8%	9.9%	30.38	30.55	183,287		
29	3.26	4.32	10.2%	10.2%	31.47	31.55	189,285		
30	3.35	4.40	10.5%	10.5%	32.55	32.58	195,480		
31	3.45	4.48	10.9%	10.9%	33.64	33.65	201,876		
32	3.54	4.56	11.2%	11.2%	34.72	34.75	208,483		
33	3.64	4.64	11.6%	11.6%	35.81	35.88	215,305		
34	3.73	4.72	11.9%	12.0%	36.89	37.06	222,351		
35	3.83	4.80	12.3%	12.3%	37.98	38.27	229,627		
36	3.92	4.88	12.6%	12.7%	39.06	39.52	237,141		
37	4.02	4.96	13.0%	13.2%	40.15	40.82	244,901		
38	4.11	5.04	13.3%	13.6%	41.23	42.15	252,916		
39	4.21	5.12	13.7%	14.0%	42.32	43.53	261,192		
40	4.30	5.20	14.0%	14.5%	43.40	44.96	269,739		
41	4.40	5.28	14.4%	15.0%	44.49	46.43	278,566		
42	4.49	5.36	14.7%	15.5%	45.57	47.95	287,682		
43	4.59	5.44	15.1%	16.0%	46.66	49.52	297,096		
44	4.68	5.52	15.4%	16.5%	47.74	51.14	306,818		
45	4.78	5.60	15.8%	17.0%	48.83	52.81	316,859		
46	4.87	5.68	16.1%	17.6%	49.91	54.54	327,228		
47	4.97	5.76	16.5%	18.2%	51.00	56.32	337,936		
48	5.06	5.84	16.8%	18.8%	52.08	58.17	348,994		
49	5.16	5.92	17.2%	19.4%	53.17	60.07	360,415		
50	5.25	6.00	17.5%	20.0%	54.25	62.03	372,209		

Used for Charts									
Equal Part	Vmax	Y	Vmax	btu/h					
EQ %	EQ %	2.00	EQ%						
		10.00	310						
In %	In V	In V	g/min	btu/h					
4.0%	0.00	2.00	0.00	0					
4.1%	0.00	2.08	0.00	0					
4.3%	0.00	2.16	0.00	0					
4.4%	0.00	2.24	0.00	0					
4.5%	0.00	2.32	0.00	0					
4.7%	0.00	2.40	0.00	0					
4.9%	0.00	2.48	0.00	0					
5.0%	15.54	2.56	15.54	93,210					
5.2%	16.04	2.64	16.04	96,260					
5.3%	16.57	2.72	16.57	99,410					
5.5%	17.11	2.80	17.11	102,663					
5.7%	17.67	2.88	17.67	106,023					
5.9%	18.25	2.96	18.25	109,492					
6.1%	18.85	3.04	18.85	113,076					
6.3%	19.46	3.12	19.46	116,776					
6.5%	20.10	3.20	20.10	120,597					
6.7%	20.76	3.28	20.76	124,544					
6.9%	21.44	3.36	21.44	128,619					
7.1%	22.14	3.44	22.14	132,828					
7.4%	22.86	3.52	22.86	137,175					
7.6%	23.61	3.60	23.61	141,664					
7.9%	24.38	3.68	24.38	146,300					
8.1%	25.18	3.76	25.18	151,087					
8.4%	26.01	3.84	26.01	156,031					
8.7%	26.86	3.92	26.86	161,137					
8.9%	27.74	4.00	27.74	166,410					
9.2%	28.64	4.08	28.64	171,856					
9.5%	29.58	4.16	29.58	177,480					
9.9%	30.55	4.24	30.55	183,287					
10.2%	31.55	4.32	31.55	189,285					
10.5%	32.58	4.40	32.58	195,480					
10.9%	33.65	4.48	33.65	201,876					
11.2%	34.75	4.56	34.75	208,483					
11.6%	35.88	4.64	35.88	215,305					
12.0%	37.06	4.72	37.06	222,351					
12.3%	38.27	4.80	38.27	229,627					
12.7%	39.52	4.88	39.52	237,141					
13.2%	40.82	4.96	40.82	244,901					
13.6%	42.15	5.04	42.15	252,916					
14.0%	43.53	5.12	43.53	261,192					
14.5%	44.96	5.20	44.96	269,739					
15.0%	46.43	5.28	46.43	278,566					
15.5%	47.95	5.36	47.95	287,682					
16.0%	49.52	5.44	49.52	297,096					
16.5%	51.14	5.52	51.14	306,818					
17.0%	52.81	5.60	52.81	316,859					
17.6%	54.54	5.68	54.54	327,228					
18.2%	56.32	5.76	56.32	337,936					
18.8%	58.17	5.84	58.17	348,994					
19.4%	60.07	5.92	60.07	360,415					
20.0%	62.03	6.00	62.03	372,209					

Projected Flow Calculations

Example:
An Energy Valve with a V'max of 310 GPM
At 2.88 volts, you would be flowing 17.67 GPM
At 6% Valve Opening
Producing 106,023 btu's at a 12 DeltaT

Hint: Refer to the Blue Columns

Figure 7 – The top of the Flow Curves tab.

5.2 Charts

The same relationships plotted: percent of valve open to GPM, DDC voltage to GPM, DDC voltage to GPM and BTU/h together, and GPM to BTU/h (a straight line, because heat transfer is flow × Delta-T × 500 at a fixed Delta-T).

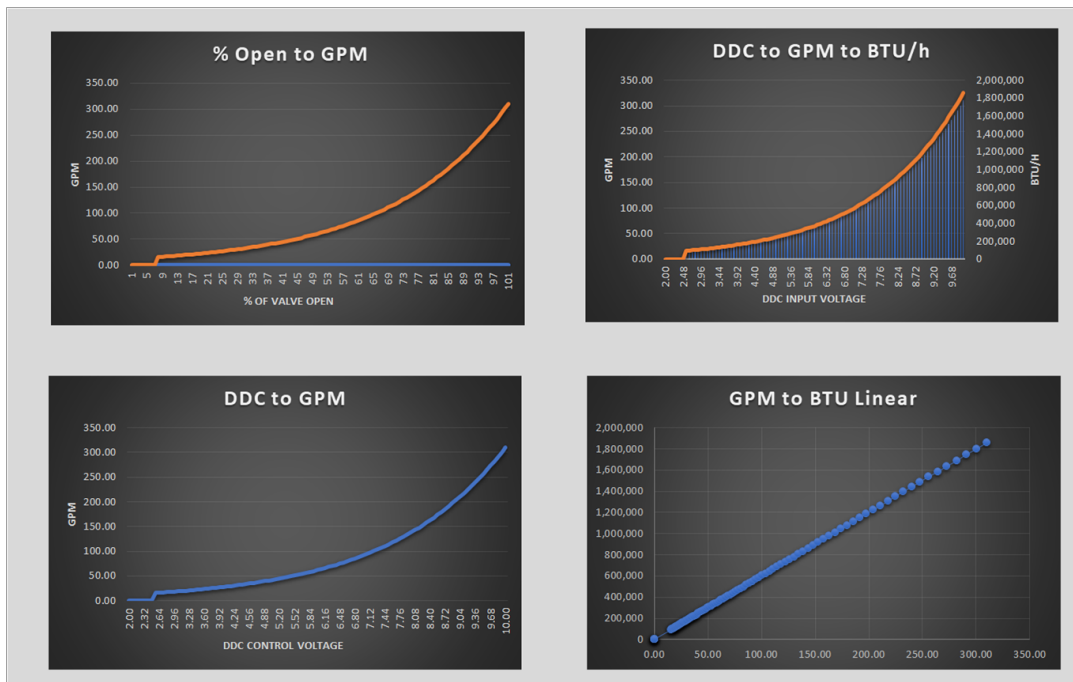


Figure 8 – The Charts tab.

6. The Results Tab

The customer-facing page. Nothing to fill in – this is what you present or print. Before and after, the full Delta-T range colour-graded from worst to best, and an interactive slider beneath.

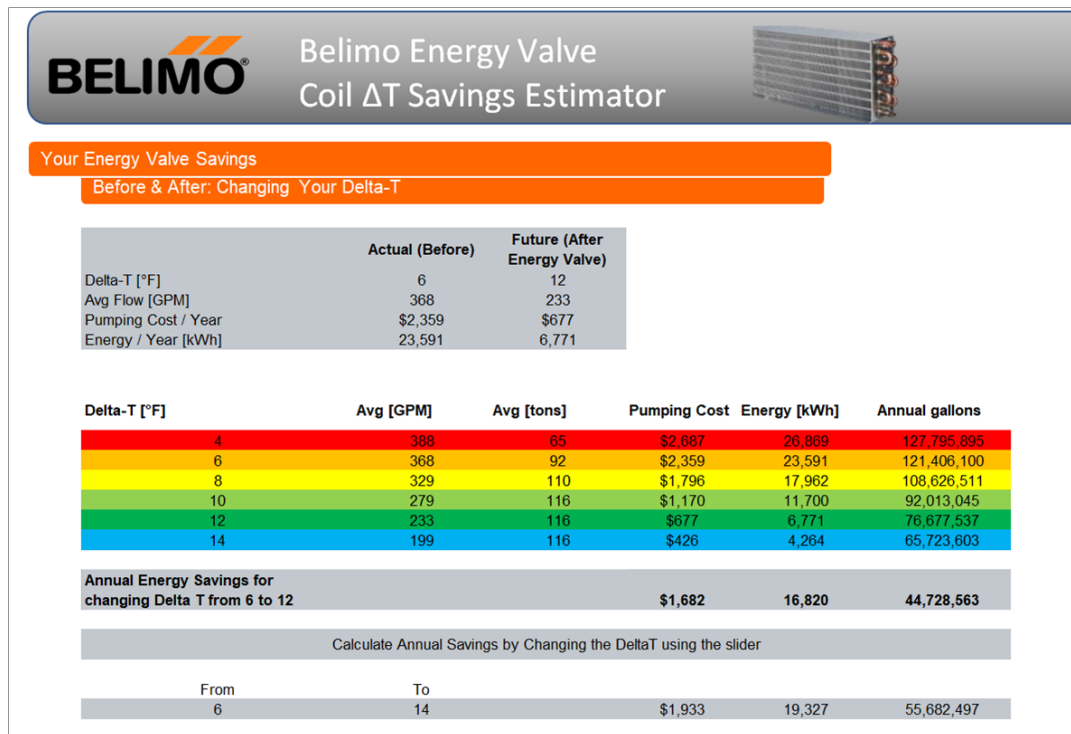


Figure 9 – The top of the Results tab: before and after, the Delta-T scenarios and the slider.

6.1 What each block shows

Block	Contents
Before & After	Your Actual and Future Delta-T side by side: Delta-T, average flow, pumping cost per year and energy per year.
Delta-T table	Six scenarios (4–14 $^{\circ}$ F) colour-graded red (worst) to blue (best), with average flow, average tonnage, pumping cost, energy and annual water volume.
Annual Energy Savings	The headline line: dollars, kilowatt-hours and gallons saved by moving from your Actual Delta-T to your Future one.
The slider	Set any “To” Delta-T with the spin buttons and the savings line beneath recalculates live, moving 2 degrees at a time. The “From” case stays pinned to the coil’s Actual Delta-T.

The slider is the most persuasive thing on the tab: it answers “what if we only get halfway there?” while the customer watches, instead of forcing a single optimistic number. Its (i) icon explains that the slider changes the Delta-T savings by 2 degrees at a time.

6.2 Greenhouse gas equivalents and financial summary

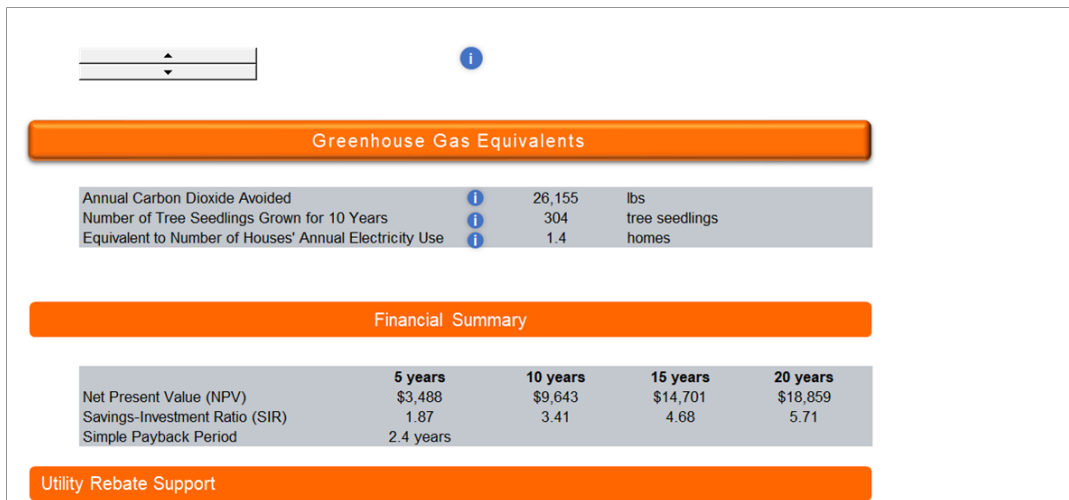


Figure 10 – The slider control, greenhouse gas equivalents and financial summary.

The greenhouse-gas block expresses the saving as pounds of carbon dioxide avoided and as two equivalents people can picture – tree seedlings grown for ten years, and a number of homes' annual electricity use. The (i) icons note that the calculations follow EPA methods; the homes figure is the energy saved divided by 12,258 kWh, the annual electricity use of a home used by the tool. The financial summary repeats the three results from the Home tab at 5, 10, 15 and 20 years.

6.3 Utility rebate support

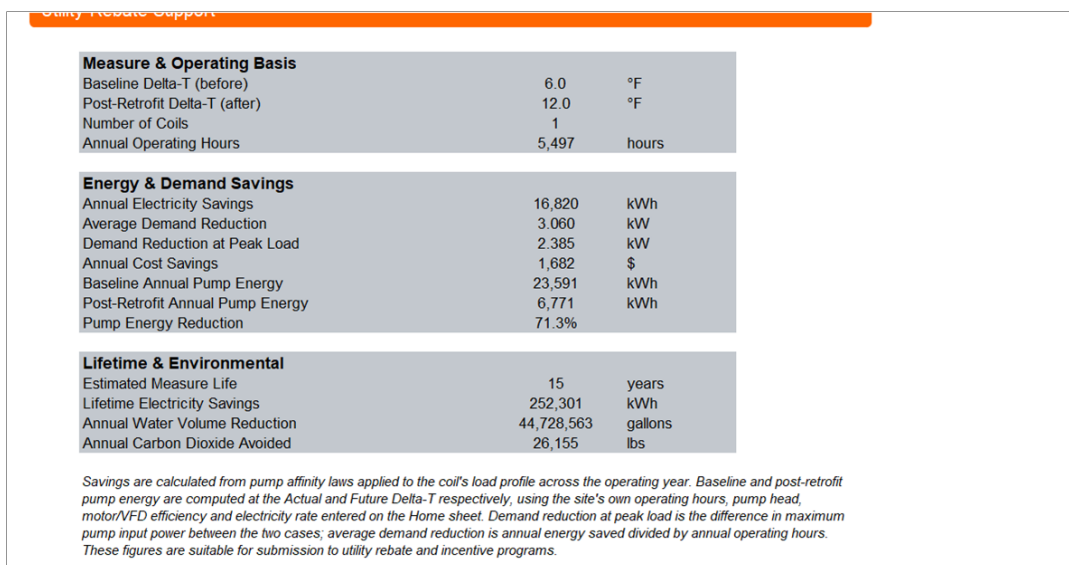


Figure 11 – The Utility Rebate Support block.

This block restates the result in the shape a utility programme wants: the measure basis (baseline and post-retrofit Delta-T, coil count, annual hours), the energy and demand savings, and the lifetime and environmental figures. It closes with a methodology note, so a reviewer can see the calculation basis without having to ask.

Two demand figures, and they are not the same thing

Average Demand Reduction is the annual energy saving divided by annual hours. **Demand Reduction at Peak Load** is the difference in pump power at the design point. A rebate form usually wants the peak figure; do not quote one where the other is asked for.

7. The Financial Results

Three numbers, shown at four horizons. These are the ones that get misread, so they are worth a careful read.

7.1 First, the idea behind all three: present value

Money in the future is worth less than money now – you could have invested today's dollar in the meantime. "Present value" is what a future amount is worth today, once you have discounted it back at the rate you set in Step 5. At a 4% discount rate, a dollar saved ten years from now counts as about 68 cents today. The discount rate is a lever: raise it and future savings count for less, so both Net Present Value and the Savings-Investment Ratio get smaller. It is not a fudge factor – set it to what your organisation actually requires and leave it there.

7.2 Net Present Value (NPV)

Take every year's saving, discount it back to today, add them up, and subtract what you spent on day one. What is left is the value the project adds, expressed in today's dollars. **Above zero** means worth doing; **below zero** means not at this discount rate – which does not mean the project loses money, only that it does not clear the return you demanded of it. It is shown at 5, 10, 15 and 20 years and grows with the horizon, because the spend happens once at the start while the savings keep accruing.

7.3 Savings-Investment Ratio (SIR)

The present value of everything you get back, divided by what you put in. Because it is a ratio rather than an amount, it lets you compare projects of very different sizes on equal terms. **Above 1.0** returns more than it costs; **below 1.0** does not pay back in present-value terms. An SIR of 1.9 means every dollar invested returns about \$1.90 in today's money. Many public-sector and utility incentive programmes set a minimum ratio as an eligibility condition, which is why it sits beside NPV on both the Home and Results tabs.

7.4 Simple Payback Period

How long the savings take to repay the investment, ignoring the time value of money entirely: the investment divided by the annual cost saving, shown in years. It is the number a building owner asks for first, and the easiest to explain. Read it as a sanity check, not as the whole case – it ignores everything that happens after payback, so it systematically undersells a long-lived measure. Quote it because people ask for it, then show the Net Present Value.

Which one do you lead with?

For a building owner, Simple Payback then Net Present Value. For a finance department, Net Present Value. For a utility or public-sector programme, the Savings-Investment Ratio – it is often written straight into the eligibility rules.

7.5 Reading the four horizons

Each result is repeated at 5, 10, 15 and 20 years, answering the same question over a different holding period: if we judged this on an N-year view, how would it look? Short horizons always look worse, because the money goes out on day one while the savings accumulate. Quote the horizon that matches how the customer actually evaluates capital – and say which horizon you are quoting.

7.6 Worked example (the values the workbook opens with)

Item	Value
Coil	310 GPM, 12 °F design Delta-T (155 tons), 30 ft head, 70% efficiency, electricity \$0.10/kWh
Operating time	20 h/day, 6 days/week, 52 weeks/year, North East (multiplier 0.88) = 5,497 hours per year
Actual → Future Delta-T	6 °F → 12 °F
Pumping cost before / after	\$2,359 per year → \$677 per year (23,591 kWh → 6,771 kWh, a 71.3% reduction)
Annual saving	\$1,682 and 16,820 kWh, and 44,728,563 gallons less water pumped
Investment / discount rate / life	\$4,000 / 4% / 15 years
NPV at 5 / 10 / 15 / 20 years	\$3,488 / \$9,643 / \$14,701 / \$18,859
SIR at 5 / 10 / 15 / 20 years	1.87 / 3.41 / 4.68 / 5.71
Simple payback	2.4 years

8. How the Numbers Are Calculated

For anyone who wants to check the arithmetic. All calculations are done in US units and converted for display.

Quantity	Calculation
Design tonnage	Design flow (GPM) × Design Delta-T ÷ 24
Flow needed at a load point and Delta-T	Design tonnage × 24 × (load %) ÷ Delta-T, limited to the pump's maximum flow (design flow × Pump Design Factor). In the design case the flow is simply load % × design flow.
Pump head at that flow	Design head × (flow ÷ design flow) ² (the pump affinity law; never below Minimum Pump head)
Pump power (HP)	Flow (GPM) × head (ft) ÷ (3960 × efficiency)
Energy (kWh)	Hours in the bin × HP × 0.746
Cost (\$)	kWh × electricity cost
Annual water volume (gal)	Flow × 60 × hours in the bin
Annual operating hours	Hours/day × days/week × weeks/year × climate multiplier
Climate multiplier	(Region degree days ÷ average degree days) × (1 + region humidity adjustment)
Annual saving	Annual pumping cost (or kWh, or gallons) at Actual Delta-T minus the same at Future Delta-T
Average demand reduction (kW)	Annual kWh saved ÷ annual operating hours
NPV (N years)	– Investment + annual cost saving × $[1 - (1 + r)^{-N}] ÷ r$ (r = discount rate; N × saving if r = 0)
SIR (N years)	Annual cost saving × $[1 - (1 + r)^{-N}] ÷ r ÷$ Investment
Simple payback	Investment ÷ annual cost saving
CO₂ avoided (lb)	kWh saved × 1.555
Tree seedlings	CO ₂ avoided converted to metric tons ÷ 0.039 tons per seedling grown for 10 years

The four load points (Load 100/75/50/25%) and their shares of the year come from Steps 3 and 4. The pump is assumed to be a single pump sized to meet and exceed the coil's design flow by the Pump Design Factor, as stated on the About screen.

9. Practicalities and Troubleshooting

9.1 Presenting and printing

The Results tab is the page to show. Print or export it once the inputs are settled – it carries the comparison, the savings, the financial summary and the rebate block on one sheet. To save a sheet as a PDF from the Pump Savings tab, use its PDF button (Section 4.4); on other tabs use Excel's File › Export or Print.

9.2 Language

Switching language on the Home tab re-labels the whole workbook, including the Results tab headings, the help messages and the chart legends. Numbers and units follow the separate Units drop-down. Your language choice is applied each time the workbook opens.

9.3 If something looks wrong

Symptom	Check
A cell will not let you type.	It is a calculated result. Compare it against the shading key in Section 2.4.
Savings look implausibly large.	Check the gap between Actual and Future Delta-T, and the Pump Design Factor on the Pump Savings tab.
The financial results look wrong.	Almost always the investment figure. Confirm it holds the incremental cost of the Energy Valve over the valve that would have been fitted anyway.
Delta-T entries are rejected.	The tool enforces a minimum gap of 2 degrees between Actual and Future, and a sensible range on Design. Adjust the pair together.
A box says "The results compare six fixed Delta T scenarios".	You typed an Actual or Future Delta-T that is not one of the six scenarios (4, 6, 8, 10, 12, 14 °F). The tool adjusted it to the nearest one and the Home and Results tabs use that value (Section 3.1).
The operating-time total shows an error.	The shares for points A, B and C in Step 3 add up to more than 100%. Reduce one of them.
Numbers did not change after an edit.	Click away from the cell to commit the entry, or press F9 to recalculate.
Buttons and the About screen do nothing.	Macros are disabled. Click Enable Editing / Enable Content (Section 2.1).
Check for Updates reports a problem.	The update source could not be reached. The tool keeps working; try again later.

This tool produces an estimate. Results depend on the accuracy of the inputs and on conditions specific to the installation. Questions, corrections and feature requests go to the maintainer named on the About screen.

Appendix. Release History

Release	Date	Changes
4.3	2026-09-24	When a newer release is published, the tool now tells you when you open it, once for each new version (Yes installs it, No carries on). Check for Updates on the About screen is always available. Fixed: on a PC whose regional settings use a comma for decimals (for example Spanish, Portuguese or French), Check for Updates could keep offering the version you already had.
4.2	2026-09-24	Added the standard admin keys used in every Belimo tool: F12 asks for the admin password and then shows and unprotects every sheet; F11 puts every sheet back exactly as it was. One password now protects every sheet in every tool. These keys are for maintainers and do not change how you use the tool. Corrected three spelling mistakes. Operating Hours / Day and Operating Days / Week no longer accept 0 (it produced #DIV/0! errors on the Pump Savings sheet). The workbook's file name no longer ends in _ML (Belimo Energy Valve Coil Estimator.xlsm); automatic updates fetch it under the new name.
4.1	2026-09-23	Actual and Future Delta-T are now matched to the six scenarios the results use (an entry such as 7 or 13 is adjusted and you are told; an Actual below 4 no longer shows errors). The Min load point now counts its hours. The homes-equivalent figure uses the 12,258 kWh per home stated in its help text. Removed a stray period from the savings heading on the Results tab.
4.0	2026-09-22	Added automatic update checking through Belimo's update source. Added Check for Updates and Download Fresh Copy buttons to the About screen. Updates save through a normal Windows Save As dialog and never touch your current copy. The About screen's version line is now live. The About screen also has a Download User Manual button, which fetches this manual.
Earlier	–	Multi-language (English, Español, Português, Français) and US/Metric units; the Home tab reorganised around a Step 1 for language and units; the consolidated Results tab with the slider, greenhouse-gas equivalents, financial summary and Utility Rebate Support block.